

问题二计算交付预览

本文件用于核对五点图及逐例附录排版，不是完整参赛论文。

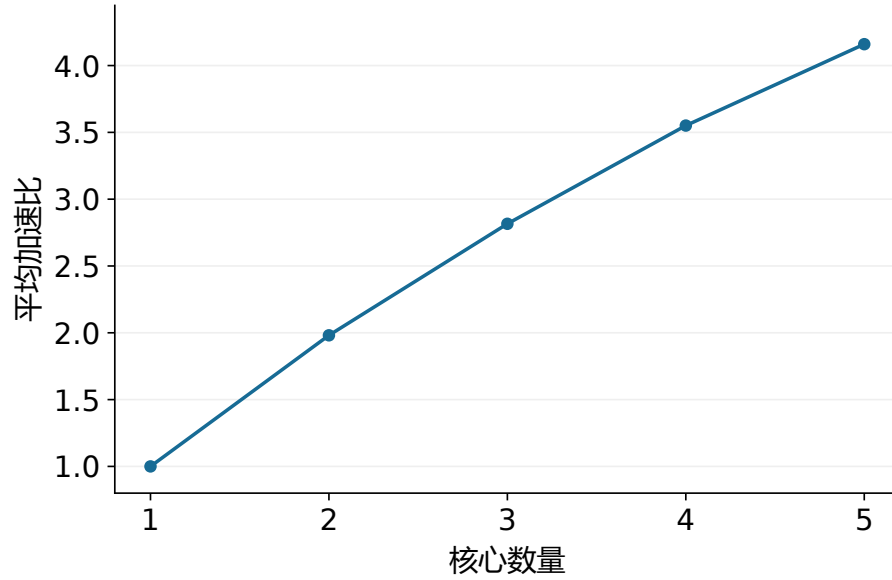


图 1: 场景 B 下 100 个算例的平均加速比，按规定单核基准与各核执行周期之比逐例取算术平均。

表 1: 问题二逐例执行周期与额外搬运量

算例	核数	执行周期	额外搬运（字节）	加速比
1	1	233110	0	1.000000
1	2	116868	1152	1.994644
1	3	78545	2304	2.967853
1	4	58984	3456	3.952089
1	5	47502	4608	4.907372
2	1	261945	0	1.000000
2	2	134224	216576	1.951551
2	3	90565	423936	2.892343
2	4	68574	377856	3.819888
2	5	55189	514560	4.746326
3	1	602212	0	1.000000
3	2	341181	6564470	1.765081
3	3	274348	9497892	2.195066
3	4	217774	7333194	2.765307
3	5	199233	7908308	3.022652
4	1	392357	0	1.000000
4	2	199074	110592	1.970910
4	3	134043	221184	2.927098
4	4	104188	331776	3.765856
4	5	88341	442368	4.441392
5	1	95618	0	1.000000
5	2	61371	882310	1.558032
5	3	54267	1461452	1.761992
5	4	52462	1495922	1.822614
5	5	50673	1688564	1.886961
6	1	82837	16400	1.000000
6	2	40815	17280	2.029572
6	3	30449	118176	2.720516
6	4	26843	203696	3.085981
6	5	27216	219928	3.043688
7	1	1012562	0	1.000000
7	2	507494	27648	1.995220
7	3	338094	55296	2.994913
7	4	254398	82944	3.980228

算例	核数	执行周期	额外搬运（字节）	加速比
7	5	203834	110592	4.967581
8	1	487605	0	1.000000
8	2	244266	0	1.996205
8	3	163359	0	2.984868
8	4	123060	0	3.962335
8	5	100603	0	4.846824
9	1	153261	0	1.000000
9	2	90801	547342	1.687878
9	3	72361	737384	2.118006
9	4	62725	838982	2.443380
9	5	56088	898504	2.732510
10	1	86969	0	1.000000
10	2	42106	311944	2.065478
10	3	30856	48384	2.818544
10	4	26340	483232	3.301784
10	5	22006	562432	3.952059
11	1	184697	0	1.000000
11	2	92646	180224	1.993578
11	3	64180	1660928	2.877797
11	4	46880	540672	3.939782
11	5	46194	2033664	3.998290
12	1	59088	0	1.000000
12	2	25535	94960	2.314000
12	3	18128	189892	3.259488
12	4	14948	39168	3.952903
12	5	12519	25344	4.719866
13	1	263949	0	1.000000
13	2	132198	8192	1.996619
13	3	88351	16384	2.987504
13	4	66982	24576	3.940596
13	5	54363	32768	4.855306
14	1	17698626	71010720	1.000000
14	2	8909568	69998520	1.986474
14	3	5951263	56526504	2.973928
14	4	4526637	59878296	3.909884
14	5	3663108	62443416	4.831587

算例	核数	执行周期	额外搬运（字节）	加速比
15	1	177362	180256	1.000000
15	2	87493	109824	2.027156
15	3	61018	298252	2.906716
15	4	48866	365824	3.629558
15	5	42569	326144	4.166459
16	1	7715523	0	1.000000
16	2	4027025	404200	1.915936
16	3	2877849	341100	2.681003
16	4	2244367	605676	3.437728
16	5	2088965	444880	3.693467
17	1	162810	22528	1.000000
17	2	76658	87552	2.123849
17	3	49850	175104	3.265998
17	4	39668	259200	4.104316
17	5	33081	331776	4.921556
18	1	1004367	1970176	1.000000
18	2	502491	2638848	1.998776
18	3	337417	3971072	2.976634
18	4	253676	4938752	3.959251
18	5	205917	5320704	4.877533
19	1	81144	0	1.000000
19	2	38506	354360	2.107308
19	3	27716	54336	2.927695
19	4	22276	762456	3.642665
19	5	18267	255068	4.442109
20	1	532746	0	1.000000
20	2	266580	0	1.998447
20	3	177950	0	2.993796
20	4	133704	0	3.984518
20	5	107650	0	4.948871
21	1	6567937	4501504	1.000000
21	2	3174339	33017856	2.069072
21	3	2090922	34854912	3.141168
21	4	1550324	4608000	4.236493
21	5	1250168	11431936	5.253644
22	1	56780	0	1.000000

算例	核数	执行周期	额外搬运（字节）	加速比
22	2	27860	145236	2.038047
22	3	18978	13824	2.991885
22	4	14336	20736	3.960658
22	5	11479	20736	4.946424
23	1	122614	0	1.000000
23	2	60965	74016	2.011220
23	3	40682	207264	3.013962
23	4	30516	281280	4.018023
23	5	24940	280992	4.916359
24	1	2555343	0	1.000000
24	2	1335245	396856	1.913763
24	3	954537	332124	2.677050
24	4	748435	595068	3.414248
24	5	695849	432232	3.672267
25	1	4863026	0	1.000000
25	2	2431840	27648	1.999731
25	3	1622164	55296	2.997863
25	4	1217444	82944	3.994456
25	5	975282	110592	4.986277
26	1	99203	0	1.000000
26	2	51429	21936	1.928931
26	3	41688	95664	2.379654
26	4	42226	172128	2.349335
26	5	38198	835212	2.597073
27	1	2363952	1400832	1.000000
27	2	1120962	1744896	2.108860
27	3	744579	18481152	3.174884
27	4	559196	3317760	4.227412
27	5	499622	21954560	4.731481
28	1	43836040	277412256	1.000000
28	2	20517451	75738432	2.136525
28	3	13687852	44350128	3.202551
28	4	10347838	74883456	4.236251
28	5	8286707	73953792	5.289923
29	1	37555	0	1.000000
29	2	18880	16388	1.989142

算例	核数	执行周期	额外搬运（字节）	加速比
29	3	12743	0	2.947108
29	4	10168	49164	3.693450
29	5	8787	65552	4.273927
30	1	2808034	5519360	1.000000
30	2	1390496	1292288	2.019448
30	3	933720	8985600	3.007362
30	4	697351	6002688	4.026715
30	5	559882	7484416	5.015403
31	1	1414369	3491328	1.000000
31	2	712262	5250048	1.985743
31	3	473822	7544832	2.985022
31	4	360134	7808000	3.927341
31	5	284848	7931904	4.965346
32	1	41443	0	1.000000
32	2	17566	6216	2.359274
32	3	11910	12360	3.479681
32	4	10728	74980	3.863069
32	5	10414	12360	3.979547
33	1	1774196	4592640	1.000000
33	2	889132	6452256	1.995425
33	3	596821	7053888	2.972744
33	4	443447	7214688	4.000920
33	5	361176	7894656	4.912275
34	1	842996	1079832	1.000000
34	2	415251	943616	2.030088
34	3	270947	2288640	3.111295
34	4	206454	3059200	4.083215
34	5	167389	3463168	5.036149
35	1	169571	0	1.000000
35	2	93692	801138	1.809877
35	3	75717	1276568	2.239537
35	4	63767	1505978	2.659228
35	5	58308	1058710	2.908194
36	1	931510	0	1.000000
36	2	466068	1152	1.998657
36	3	311345	2304	2.991890

算例	核数	执行周期	额外搬运（字节）	加速比
36	4	233584	3456	3.987902
36	5	187182	4608	4.976493
37	1	210001	0	1.000000
37	2	105374	204800	1.992911
37	3	74038	573440	2.836395
37	4	53320	614400	3.938503
37	5	43013	942080	4.882268
38	1	1364555	5627904	1.000000
38	2	674500	6790656	2.023062
38	3	449590	8107008	3.035110
38	4	338182	8954880	4.034972
38	5	275278	9592320	4.957007
39	1	1435262	7929856	1.000000
39	2	708430	9166848	2.025976
39	3	478013	11911168	3.002559
39	4	368961	14536704	3.890010
39	5	338925	14536704	4.234748
40	1	253454	0	1.000000
40	2	134642	1227916	1.882429
40	3	105492	1985768	2.402590
40	4	89539	1836692	2.830655
40	5	78938	1915296	3.210798
41	1	6003957	30898176	1.000000
41	2	2990398	33669120	2.007745
41	3	1991911	34851840	3.014169
41	4	1490603	39164928	4.027871
41	5	1196379	40630272	5.018441
42	1	1686740	6629376	1.000000
42	2	840111	7262496	2.007758
42	3	570685	8474176	2.955641
42	4	430510	8605536	3.918004
42	5	350713	9936736	4.809460
43	1	677656	49160	1.000000
43	2	349881	3210754	1.936819
43	3	250900	4852554	2.700901
43	4	207895	5894554	3.259607

算例	核数	执行周期	额外搬运（字节）	加速比
43	5	199041	6609206	3.404605
44	1	154407	4438112	1.000000
44	2	54510	886880	2.832636
44	3	44484	891584	3.471068
44	4	49989	1389344	3.088820
44	5	43028	655488	3.588524
45	1	125614	0	1.000000
45	2	63059	0	1.992007
45	3	42601	0	2.948616
45	4	31878	0	3.940461
45	5	26164	0	4.801024
46	1	276455	3176288	1.000000
46	2	112410	942688	2.459345
46	3	96500	2958528	2.864819
46	4	71871	2098976	3.846545
46	5	77698	2660736	3.558071
47	1	326508	0	1.000000
47	2	197413	1264006	1.653934
47	3	156690	1857324	2.083783
47	4	143256	2256146	2.279192
47	5	135565	2575320	2.408498
48	1	222220	0	1.000000
48	2	150049	467846	1.480983
48	3	114626	1177612	1.938653
48	4	114384	1270930	1.942754
48	5	101641	1273560	2.186322
49	1	196276	0	1.000000
49	2	126174	1027062	1.555598
49	3	83711	2408560	2.344686
49	4	75814	2607184	2.588915
49	5	76846	2907980	2.554147
50	1	149690	0	1.000000
50	2	87435	1022694	1.712015
50	3	60421	1831624	2.477450
50	4	57912	2053712	2.584784
50	5	57017	1337182	2.625357

算例	核数	执行周期	额外搬运（字节）	加速比
51	1	607628	0	1.000000
51	2	319230	394084	1.903418
51	3	228581	328736	2.658261
51	4	183794	591064	3.306027
51	5	170016	427456	3.573946
52	1	180301	0	1.000000
52	2	90903	102912	1.983444
52	3	60160	181248	2.997025
52	4	45777	254976	3.938681
52	5	37846	333312	4.764070
53	1	1472686	5405536	1.000000
53	2	723016	4631434	2.036865
53	3	495435	6090084	2.972511
53	4	394500	8679750	3.733044
53	5	357068	9823040	4.124385
54	1	815051	0	1.000000
54	2	465795	2620302	1.749806
54	3	359793	4112418	2.265333
54	4	274779	7279264	2.966206
54	5	257061	8197794	3.170652
55	1	118835	69632	1.000000
55	2	52737	16384	2.253352
55	3	35943	74240	3.306207
55	4	31281	115712	3.798951
55	5	24831	173568	4.785752
56	1	277285	0	1.000000
56	2	172836	3326338	1.604324
56	3	122295	3278908	2.267345
56	4	115731	3685886	2.395944
56	5	117134	4124830	2.367246
57	1	59726	94216	1.000000
57	2	29703	22304	2.010773
57	3	20565	14112	2.904255
57	4	16345	28224	3.654084
57	5	14028	14400	4.257628
58	1	4539717	28696576	1.000000

算例	核数	执行周期	额外搬运（字节）	加速比
58	2	2266495	29548544	2.002968
58	3	1538252	30400512	2.951218
58	4	1141935	31551488	3.975460
58	5	1000800	32104448	4.536088
59	1	829473	1691648	1.000000
59	2	369318	360448	2.245959
59	3	254073	1171456	3.264703
59	4	185216	1081344	4.478409
59	5	150782	2072576	5.501141
60	1	888789	2068520	1.000000
60	2	441525	2746508	2.012998
60	3	295983	3159296	3.002838
60	4	223107	3960200	3.983689
60	5	180081	3733888	4.935496
61	1	77191	0	1.000000
61	2	39298	34560	1.964248
61	3	28342	43008	2.723555
61	4	23892	62208	3.230830
61	5	18840	110592	4.097187
62	1	2854809	0	1.000000
62	2	1431735	2347008	1.993951
62	3	958567	2268672	2.978205
62	4	722454	2603520	3.951544
62	5	579973	2746368	4.922314
63	1	1070397	0	1.000000
63	2	536070	1049088	1.996749
63	3	358763	1262592	2.983577
63	4	270138	1420800	3.962408
63	5	218917	1539072	4.889511
64	1	19116	0	1.000000
64	2	14130	47620	1.352866
64	3	13074	41446	1.462139
64	4	10160	130956	1.881496
64	5	11391	159314	1.678167
65	1	53068	0	1.000000
65	2	25517	13968	2.079712

算例	核数	执行周期	额外搬运（字节）	加速比
65	3	17964	27936	2.954130
65	4	13800	41904	3.845507
65	5	11212	55872	4.733143
66	1	359070	0	1.000000
66	2	203504	1653418	1.764437
66	3	146357	2500738	2.453385
66	4	116226	3806464	3.089412
66	5	106067	3979388	3.385313
67	1	63834834	264449408	1.000000
67	2	32040130	215994240	1.992340
67	3	21255269	159238912	3.003248
67	4	16409509	226600576	3.890112
67	5	13461153	229484544	4.742152
68	1	362095	0	1.000000
68	2	196346	1624576	1.844168
68	3	180284	2355384	2.008470
68	4	156990	2646854	2.306484
68	5	158423	5510878	2.285621
69	1	25144	0	1.000000
69	2	17030	364198	1.476453
69	3	14179	325098	1.773327
69	4	15620	361896	1.609731
69	5	13147	433008	1.912528
70	1	117300	0	1.000000
70	2	54493	129536	2.152570
70	3	39333	259072	2.982229
70	4	30080	333312	3.899601
70	5	25374	449024	4.622842
71	1	18919	0	1.000000
71	2	12313	171622	1.536506
71	3	9867	250696	1.917401
71	4	10732	223406	1.762859
71	5	10542	307214	1.794631
72	1	23897828	108613680	1.000000
72	2	11836023	113724192	2.019076
72	3	7941294	120851232	3.009312

算例	核数	执行周期	额外搬运（字节）	加速比
72	4	5965868	123071264	4.005759
72	5	4826156	77245216	4.951731
73	1	10081626	141570432	1.000000
73	2	5294770	86179712	1.904073
73	3	3805931	134138624	2.648925
73	4	3295138	137012864	3.059546
73	5	2848898	94477312	3.538781
74	1	1682239	6070280	1.000000
74	2	814840	3776512	2.064502
74	3	542996	5619712	3.098069
74	4	408908	6623232	4.113979
74	5	328499	7151616	5.120987
75	1	1441809	0	1.000000
75	2	754663	4745798	1.910534
75	3	557783	12507404	2.584892
75	4	462730	11096688	3.115875
75	5	378054	12427608	3.813765
76	1	6185113	28280832	1.000000
76	2	3065214	31095808	2.017841
76	3	2046322	17912832	3.022551
76	4	1527039	19563520	4.050396
76	5	1227673	22311936	5.038079
77	1	495320	0	1.000000
77	2	248734	57344	1.991364
77	3	168976	114688	2.931304
77	4	127832	114688	3.874773
77	5	114837	172032	4.313244
78	1	2296143	12644568	1.000000
78	2	1050081	2782080	2.186634
78	3	718909	6965760	3.193927
78	4	540759	6950784	4.246148
78	5	470359	10766304	4.881682
79	1	31052251	32448512	1.000000
79	2	15118882	194232320	2.053872
79	3	10181692	39960576	3.049812
79	4	7620379	31801344	4.074896

算例	核数	执行周期	额外搬运（字节）	加速比
79	5	6063012	34084864	5.121588
80	1	292138	548864	1.000000
80	2	145825	1675264	2.003346
80	3	100325	3047424	2.911916
80	4	90584	2666496	3.225051
80	5	89345	3047424	3.269774
81	1	151795	0	1.000000
81	2	76860	0	1.974954
81	3	51943	0	2.922338
81	4	39588	0	3.834369
81	5	30943	0	4.905633
82	1	721041	0	1.000000
82	2	407535	3389668	1.769274
82	3	288553	5338828	2.498817
82	4	235778	6420240	3.058135
82	5	206503	7708408	3.491673
83	1	1057993	13934688	1.000000
83	2	442847	5589600	2.389071
83	3	290664	5739712	3.639918
83	4	220741	6471456	4.792916
83	5	201107	7741824	5.260846
84	1	2507412	0	1.000000
84	2	1175864	32004864	2.132400
84	3	836220	0	2.998508
84	4	629028	0	3.986169
84	5	503472	0	4.980241
85	1	3187899	0	1.000000
85	2	1843091	17607566	1.729648
85	3	1276351	31680716	2.497666
85	4	995687	27902578	3.201708
85	5	890475	38398484	3.579998
86	1	118403	0	1.000000
86	2	74609	1019206	1.586980
86	3	63229	2197292	1.872606
86	4	62299	2353614	1.900560
86	5	60402	2265748	1.960250

算例	核数	执行周期	额外搬运（字节）	加速比
87	1	3974071	7902720	1.000000
87	2	1987606	10142208	1.999426
87	3	1374901	11036160	2.890442
87	4	1046846	12607488	3.796233
87	5	833349	12487680	4.768796
88	1	243623	0	1.000000
88	2	132639	1032234	1.836737
88	3	122079	1572244	1.995618
88	4	112565	2028164	2.164287
88	5	112833	2034888	2.159147
89	1	693092	0	1.000000
89	2	329513	1467520	2.103383
89	3	225452	2429184	3.074233
89	4	165365	1827200	4.191286
89	5	133148	2186320	5.205426
90	1	1178363	21233592	1.000000
90	2	470442	3196848	2.504800
90	3	342250	5523456	3.442989
90	4	294798	7360608	3.997188
90	5	267434	7981056	4.406182
91	1	9186769	53947008	1.000000
91	2	4532609	52032556	2.026817
91	3	3042680	60113508	3.019302
91	4	2314443	67112896	3.969322
91	5	1872567	71035764	4.905976
92	1	4815554	66000096	1.000000
92	2	2017778	10740320	2.386563
92	3	1348585	21856448	3.570820
92	4	1018849	18053920	4.726465
92	5	826308	19945856	5.827795
93	1	74205	0	1.000000
93	2	37827	8192	1.961694
93	3	25103	16384	2.956021
93	4	19779	24576	3.751706
93	5	15876	32768	4.674036
94	1	94892	0	1.000000

算例	核数	执行周期	额外搬运（字节）	加速比
94	2	47498	106272	1.997810
94	3	32728	212544	2.899413
94	4	24668	318528	3.846765
94	5	21804	424800	4.352045
95	1	2084853	0	1.000000
95	2	1042890	0	1.999111
95	3	695775	0	2.996447
95	4	524422	0	3.975525
95	5	420852	0	4.953886
96	1	84194	0	1.000000
96	2	42443	346520	1.983696
96	3	30876	123264	2.726843
96	4	24217	234168	3.476649
96	5	22137	448896	3.803316
97	1	11491509	10584064	1.000000
97	2	5153667	2490368	2.229773
97	3	3444086	82444288	3.336592
97	4	2622394	14221312	4.382068
97	5	2102348	21311488	5.466036
98	1	177430	0	1.000000
98	2	89620	0	1.979804
98	3	65476	784240	2.709848
98	4	53735	961274	3.301945
98	5	47355	1166870	3.746806
99	1	1622999	5484544	1.000000
99	2	807983	5906432	2.008704
99	3	540063	6479872	3.005203
99	4	405336	9838592	4.004083
99	5	332266	7084052	4.884638
100	1	151223	0	1.000000
100	2	76850	621700	1.967768
100	3	57811	72192	2.615817
100	4	49623	1207894	3.047438
100	5	53383	1599352	2.832793